(Chapter 19), respectively. Both of these one-dimensional filters can be rapidly carried out by recursion. This results in an image convolution time proportional to only N^2, completely independent of the size of the PSF. In other words, an image can be convolved with as large a PSF as needed, with only a few integer operations per pixel. For example, the convolution of a 512×512 image requires only a few hundred milliseconds on a personal computer. That's fast! Don't like the shape of these two filter kernels? Convolve the image with one of them *several times* to approximate a Gaussian PSF (guaranteed by the Central Limit Theorem, Chapter 7). These are great algorithms, capable of snatching success from the jaws of failure. They are well worth remembering.

Example of a Large PSF: Illumination Flattening

A common application requiring a large PSF is the enhancement of images with unequal illumination. Convolution by separability is an ideal algorithm to carry out this processing. With only a few exceptions, the images seen by the eye are formed from *reflected* light. This means that a viewed image is equal to the reflectance of the objects multiplied by the ambient illumination. Figure 24-8 shows how this works. Figure (a) represents the *reflectance* of a scene being viewed, in this case, a series of light and dark bands. Figure (b) illustrates an example illumination signal, the pattern of light falling on (a). As in the real world, the illumination slowly varies over the imaging area. Figure (c) is the image seen by the eye, equal to the reflectance image, (a), multiplied by the illumination image, (b). The regions of poor illumination are difficult to view in (c) for two reasons: they are too dark and their contrast is too low (the difference between the peaks and the valleys).

To understand how this relates to the problem of every day vision, imagine you are looking at two identically dressed men. One of them is standing in the bright sunlight, while the other is standing in the shade of a nearby tree. The percent of the incident light reflected from both men is the same. For instance, their faces might reflect 80% of the incident light, their gray shirts 40% and their dark pants 5%. The problem is, the illumination of the two might be, say, ten times different. This makes the image of the man in the shade ten times darker than the person in the sunlight, and the contrast (between the face, shirt, and pants) ten times less.

The goal of the image processing is to *flatten* the illumination component in the acquired image. In other words, we want the final image to be representative of the objects' reflectance, not the lighting conditions. In terms of Fig. 24-8, given (c), find (a). This is a nonlinear filtering problem, since the component images were combined by multiplication, not addition. While this separation cannot be performed perfectly, the improvement can be dramatic.

To start, we will convolve image (c) with a large PSF, one-fifth the size of the entire image. The goal is to eliminate the sharp features in (c), resulting